



High-order immersed boundary method for inviscid flows applied to flux reconstruction method on a hierarchical Cartesian grid

Masaya Funada^{a,*}, Taro Imamura^{a,1}

^aThe university of Tokyo, Department of Aeronautics and Astronautics, 7-3-1, Hongo, Bunkyo, Tokyo 113-8656, Japan

ARTICLE INFO

Article history:

Keywords: immersed boundary method, flux reconstruction method, hierarchical Cartesian grid

ABSTRACT

We propose novel high-order immersed boundary methods for simulating inviscid flows on hierarchical Cartesian grids using the flux reconstruction (FR) method. A high-order distribution of physical quantities derived from the FR method is utilized to interpolate flow variables to maintain the same order near the wall as the interior domain. In particular, the wall's curvature effect is investigated in depth. We validate the accuracy through basic 2-D and 3-D simulations of inviscid flows. Consequently, the same accuracy as the order of the FR method (up to the fourth order) is achieved, including in the near-wall region.

© 2023 Elsevier Inc. All rights reserved.

1. Introduction

The aerodynamic design of aircraft depends heavily on computational fluid dynamics. One of the essential objectives in this field is to improve the accuracy of unsteady turbulent flow prediction, particularly in aircraft high-lift devices [1]. In recent years, large-eddy simulation (LES), particularly wall-modeled LES, has attracted attention for accurately analyzing such unsteady turbulent flow [2]. The discontinuous Galerkin (DG) method [3], the spectral difference (SD) method [4], and the flux reconstruction (FR) method [5] have been proposed as high-order accurate methods applicable to such problems. These methods apply to unstructured grids. In addition, these methods can achieve a high parallelization efficiency because only neighboring cells are required to update the physical quantities within a cell.

Hierarchical Cartesian grids have the advantage of automatic and robust grid generation for complex geometries [6–8]. Flow around complex geometries such as a landing gear [9] and an aircraft configuration, NASA Common Research Model [10] are solved with hierarchical Cartesian grids. The hierarchical Cartesian grid, which is

*Corresponding author. E-mail address: masa-doradora-1999@g.ecc.u-tokyo.ac.jp

not a body-fitted grid, requires efforts for near-wall treatment. The immersed boundary method [11] is an example of a technique that can treat wall boundaries on hierarchical Cartesian grids without deforming the grid. Immersed boundary methods are divided into two categories [12]: the continuous forcing approach [13–15], which applies a volumetric force to the region in which an object resides, and the discrete forcing approach [16–21], which imposes wall boundary conditions directly on the discretized governing equations. In the continuous forcing approach, interpolating physical quantities on the wall surface is not required, and application to moving objects is straightforward [13, 14]. The disadvantage is that it is difficult to resolve the thin turbulent wall boundary layer due to the finite thickness of the region where volume forces are imposed in the flow, particularly for high Reynolds number flows. In contrast, the discrete forcing approach can sharply capture the wall boundary while interpolating physical quantities from the fluid region to the wall is necessary.

Immersed boundary methods are typically accurate to the second or lower order. Kou et al. [15] applied an immersed boundary method based on the continuous forcing approach to the FR method and achieved fifth-order spatial accuracy in the region far from the wall and first-order spatial accuracy in the near-wall region for the advection-diffusion problem. Palma et al. [16] and Tullio et al. [17] used linear interpolation with finite volume methods to achieve first-order or second-order accuracy for compressible viscous flows. Some studies have proposed higher-order-immersed boundary methods. With finite difference methods, polynomial interpolation of physical quantities using multiple points in fluid cells [18, 19] achieved spatial accuracy exceeding the fourth-order for the wave equation, linear advection equation, and Burger's equation. With finite difference methods, weighted least-squares interpolation [20, 21] also achieved spatial accuracy exceeding the third-order for the advection-diffusion and acoustic wave scattering problems. By combining polynomial interpolation and weighted least-squares interpolation, Khalili et al. [22] achieved superior spatial accuracy to the third-order for compressible viscous flow with the finite difference method. However, numerous stencils are required for the interpolation of the physical quantities and some points are located far from the wall.

Additionally, when simulating inviscid flows around curved bodies, it is necessary to account for the curvature of the wall. Dadone et al. [23] proposed the curvature corrected symmetry technique (CCST), which modifies the wall boundary conditions for pressure and density gradients to account for the curvature of the wall. CCST has enhanced the accuracy of simulations on a body-fitted grid compared to the zeroth order or first-order pressure extrapolation techniques. In addition to pressure and density gradients, Fidalgo et al. [24] modified the boundary conditions for the velocity component tangential to the wall to account for curvature, achieving fifth-order spatial accuracy in numerical simulations with a body-fitted grid for flow around an inviscid cylinder. Zhang et al. [25] attained second-order spatial accuracy on a cartesian grid by applying CCST to the immersed boundary method.

This study aims to develop immersed boundary methods for inviscid flow simulation on hierarchical Cartesian grids capable of performing high-order accurate analysis even in the region close to the wall. The FR method is utilized as a high-order scheme, and the discrete forcing approach is adopted. The adiabatic solid wall boundary conditions for inviscid flow simulations are proposed, and the effect of the wall curvature is investigated in detail. Using fundamental 2-D and 3-D inviscid flow simulations, we demonstrate that the same accuracy as the order of the

FR method can be achieved even in the near-wall region.

The paper is structured as follows: Section 2 describes the FR method, the governing equations, the treatment of wall boundary conditions, and the time integration method. In Section 3, several simulations of inviscid flow are conducted to validate the present techniques. The fourth section concludes this paper.

2. Numerical methods

2.1. Flux Reconstruction Method

This section presents an overview of the FR method [5]. For brevity, we consider the following one-dimensional advection equation:

$$\frac{\partial q}{\partial t} + \frac{\partial f(q)}{\partial x} = 0, \quad (1)$$

where q denotes the conservative variable, $f(q)$ denotes the flux, t denotes the time, and x denotes the one-dimensional coordinate. First, the x -coordinates of the cells in the computational domain are transformed to the generalized coordinates ξ , from -1 to 1 , that the following relationship is satisfied:

$$x = \frac{1 - \xi}{2} x_{i-1/2} + \frac{1 + \xi}{2} x_{i+1/2}, \quad (2)$$

where i denotes the index of each cell, and $x_{i-1/2}$ and $x_{i+1/2}$ denote the x -coordinates of cell interfaces. Then, k solution points (SPs) are placed in each cell. SPs are positioned at the Gauss points in this study. On each SP, a physical quantity is specified, and the distribution of the conservative variable is approximated by a polynomial of order $k - 1$ as

$$q_i(\xi) = \sum_{j=1}^k q_{i,j} \phi_j(\xi), \quad (3)$$

where j denotes the index of SPs, $q_{i,j}$ denotes the conservative variable of the j th SP in cell i , and $\phi_j(\xi)$ denotes a $(k - 1)$ th-order function expressed as

$$\phi_j(\xi) = \prod_{m=1, m \neq j}^k \frac{\xi - \xi_m}{\xi_j - \xi_m}, \quad (4)$$

where ξ_j denotes the ξ -coordinate of the j th SP. To determine the time evolution at each SP, we determine the spatial derivative of the flux f . The flux f is expressed in the same way as the conservative variable q as follows:

$$f_i(\xi) = \sum_{j=1}^k f_{i,j} \phi_j(\xi). \quad (5)$$

We can obtain the spatial derivative of flux f by differentiating Eq. (5) as follows:

$$\frac{\partial f_i}{\partial x} = \frac{\partial \xi}{\partial x} \frac{\partial f_i}{\partial \xi} = \frac{2}{\Delta x_i} \sum_{j=1}^k f_{i,j} \phi'_j(\xi), \quad (6)$$

where Δx_i denotes the grid width of cell i , and the dash symbol denotes the derivative of a function. However, this derivative of the flux is discontinuous at the cell interfaces because the resulting function only utilizes information within a given cell and does not account for interactions with neighboring cells. We modify the flux in Eq. (5) such

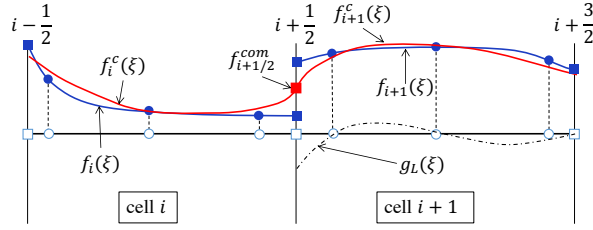


Fig. 1: Schematic of the flux reconstruction procedure.

that the fluxes are the same on both sides of the cell interface. If f_i^c is the corrected flux and σ_i is the correction term of the flux, then the following relationship holds:

$$f_i^c(\xi) = f_i(\xi) + \sigma_i(\xi). \quad (7)$$

From Eq. (1), the spatial derivative of the corrected flux f_i^c must be of order $k - 1$, the same as the polynomial of the conservative variable q ; therefore, σ_i is a k th-order polynomial. Next, the σ_i is divided into contributions from the left and right interfaces of the cell expressed as follows:

$$\sigma_i(\xi) = [f_{i-1/2}^{com} - f_i(-1)]g_L(\xi) + [f_{i+1/2}^{com} - f_i(1)]g_R(\xi), \quad (8)$$

where $f_i(-1)$ and $f_i(1)$ denote the fluxes obtained from Eq. (5), $f_{i-1/2}^{com}$ denotes the flux common on both sides of the interface between cell $i - 1$ and cell i , and $f_{i+1/2}^{com}$ is the flux common on both sides of the interface between cell i and cell $i + 1$. The interfacial fluxes $f_{i-1/2}^{com}$ and $f_{i+1/2}^{com}$ are evaluated using SLAU [26]. The functions $g_L(\xi)$ and $g_R(\xi)$ are k th-order polynomials, called correction functions, which satisfy the following relations at $\xi = -1, 1$:

$$\begin{aligned} g_L(-1) &= 1, g_L(1) = 0, \\ g_R(-1) &= 0, g_R(1) = 1. \end{aligned} \quad (9)$$

There are some correction functions. However, in this study, we adapt g_{HU} [5], which has a large upper bound on the Courant number as g_L . The function g_{HU} is as follows:

$$g_{HU} = \frac{(-1)^k}{2} \left[\frac{(k-1)L_k + kL_{k-2}}{2k-1} - L_{k-1} \right], \quad (10)$$

where L_k denotes a k th Legendre polynomial. The function $g_R(\xi) = g_L(-\xi)$. Figure 1 demonstrates the schematic of the flux reconstruction procedure described above. Thus, the spatial derivative of the corrected flux is as

$$\frac{\partial f_i^c}{\partial \xi} = \sum_{j=1}^k f_{i,j} \phi_j'(\xi) + [f_{i-1/2}^{com} - f_i(-1)]g_L'(\xi) + [f_{i+1/2}^{com} - f_i(1)]g_R'(\xi). \quad (11)$$

The time evolution of the physical quantity at each SP can be computed using Eq. (1). The FR method can achieve k th-order accuracy using k SPs.

To execute the 2-D simulations, we transform the (x, y) coordinates to $(\xi, \eta) \in [-1, 1] \times [-1, 1]$. SPs are positioned within each cell, while flux points (FPs) are positioned at the cell interface, as shown in Figure 2. The polynomial distribution of the physical quantity within a cell is as follows:

$$q_i(\xi, \eta) = \sum_{l=1}^k \sum_{m=1}^k q_{i,l,m} \phi_l(\xi) \phi_m(\eta), \quad (12)$$

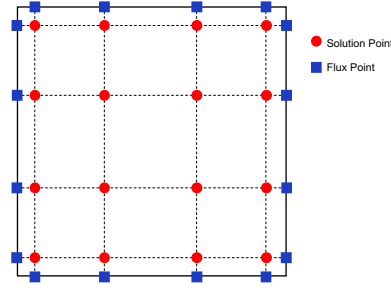


Fig. 2: Solution points and flux points in 2-D simulations.

where l and m denote the indexes of the SP in the ξ and η directions and $q_{i,l,m}$ denotes the physical quantity at the (l, m) -th SP. We reconstruct the flux by repeating the operation in the 1-D case in two directions. Then, for each FP, the common flux at the cell interface is determined. It is worth noting that in the 3-D simulations, the 1-D operation is repeated in three directions. In the hierarchical Cartesian grid, the positions of the FPs on both sides of the cell interface where the cell sizes are different do not coincide, as shown in Figure 3. Therefore, we use the mortar element method [27] to transform the physical quantities. The black, red, and blue points on each interfacial cell are FPs. The gray points represent the placement of the FPs in the larger cell in the same manner as the red and blue FPs. The procedure for transformation consists of three steps: 1) The physical quantities of the black FPs in the larger cells are projected onto the gray FPs; 2) Common fluxes are calculated between the gray FPs and red and blue FPs; 3) The computed fluxes are projected back onto the black FPs.

2.2. Governing equations

We consider the following compressible Euler equations in conservation form:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_j}(\rho u_j) = 0, \quad (13a)$$

$$\frac{\partial}{\partial t}(\rho u_i) + \frac{\partial}{\partial x_j}(\rho u_i u_j) + \frac{\partial P}{\partial x_i} = 0, \quad (13b)$$

$$\frac{\partial}{\partial t}(\rho E) + \frac{\partial}{\partial x_j}[(\rho E + P)u_j] = 0, \quad (13c)$$

where ρ denotes the density, u_i denotes the i th component of the velocity vector, P denotes the pressure, and E denotes the specific total energy.

2.3. Wall boundary conditions

Figure 4 demonstrates the classification of the cells. The cells that intersect the wall are considered wall cells. The cells outside the wall are classified as fluid cells, and the flow simulation is carried out in these cells. The cells contained within the wall are categorized as body cells. Figure 5 illustrates the schematic of the flow variables at the interface between fluid and wall cells. Herein, $\mathbf{q}_{FP}$ denotes the vector of the flow variables at the FP in the wall cell, and $\mathbf{q}_{FP,f}$ denotes the flow variables at the FP in the fluid cell. The imposed wall boundary conditions in the immersed boundary methods are as follows. First, the physical quantities $\mathbf{q}_{FP}$ are interpolated using the physical quantities in the fluid region and the wall boundary conditions, $\mathbf{q}_{FP,f}$ is then calculated from the polynomial distributions in the fluid

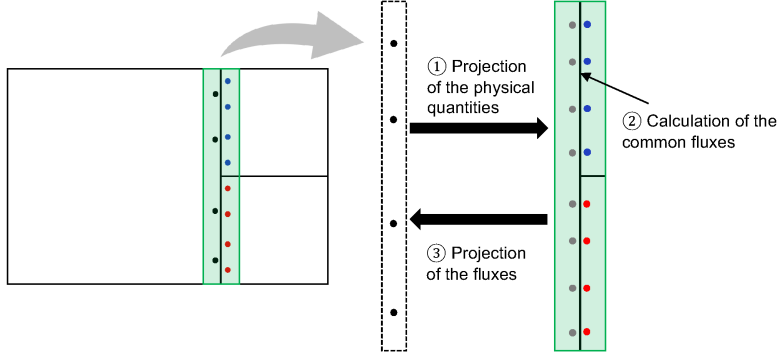


Fig. 3: Schematic of the mortar element method.

cell. Then, the flux is computed using SLAU with $\mathbf{q}_{FP}$ and $\mathbf{q}_{FP,f}$. Therefore, imposing the wall boundary conditions results in interpolating $\mathbf{q}_{FP}$.

For 2-D inviscid flow without discontinuities, two alternative sets of adiabatic slip wall boundary conditions are defined as follows [24, 28, 29]:

$$u_n = 0, \quad (14a)$$

$$\frac{\partial \rho}{\partial n} = \frac{\rho}{\gamma P} \frac{\partial P}{\partial n}, \quad (14b)$$

$$\frac{\partial P}{\partial n} = -\rho \frac{u_t^2}{R}, \quad (14c)$$

$$\frac{\partial u_t}{\partial n} = \frac{u_t}{R}, \quad (14d)$$

$$u_n = 0, \quad (15a)$$

$$\frac{\partial H}{\partial n} = \frac{\partial}{\partial n} \left(\frac{E + P}{\rho} \right) = 0, \quad (15b)$$

$$\frac{\partial S}{\partial n} = \frac{\partial}{\partial n} \left(\frac{P}{\rho^\gamma} \right) = 0, \quad (15c)$$

$$\frac{\partial u_t}{\partial n} = \frac{u_t}{R}, \quad (15d)$$

where R denotes the radius of the wall curvature, γ denotes the specific heat ratio, u_n denotes the velocity component perpendicular to the wall, u_t denotes the velocity component tangential to the wall, H denotes the specific total enthalpy, and n denotes the direction perpendicular to the wall. Next, we consider interpolation methods for physical quantities $\mathbf{q}_{FP}$ that satisfy Eqs. (14) or Eqs. (15) on a wall. We place image points (IPs) in the near-wall fluid cells to define flow variables used in the interpolation to the FPs. In the following, we describe two kinds of methods: a low-order method using a single IP and high-order methods using multiple IPs.

2.3.1. A low-order method

Figure 6 shows the schematic of the low-order wall boundary condition. An IP is placed perpendicular to the wall through the FP (blue square). Initially, the physical quantity at the IP is derived from the polynomial distribution in the

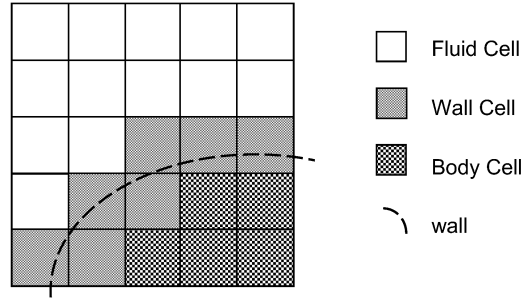


Fig. 4: Classification of cells.

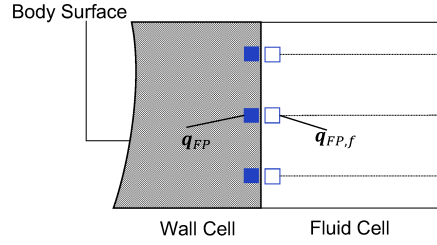


Fig. 5: Schematic of the flow variables at the interface between a fluid cell and a wall cell.

cell containing the IP using Eq. (12). Since Eqs. (14) and (15) are equivalent, either equations can be used. Equations (14) was adopted because it is expressed in terms of primitive variables, which simplifies code implementation. Using Eqs. (14), we obtain the physical quantities at the FP as follows:

$$u_{n,FP} = \frac{d_{FP}}{d_{IP}} u_{n,IP}, \quad (16a)$$

$$\rho_{FP} = \rho_{IP}, \quad (16b)$$

$$P_{FP} = P_{IP}, \quad (16c)$$

$$u_{t,FP} = u_{t,IP}, \quad (16d)$$

where the subscript FP denotes the physical quantity at the FP, and the subscript IP denotes the physical quantity at the IP. The variables d_{FP} and d_{IP} denote the distances of the FP and IP from the wall, respectively. The value d_{FP} is calculated before flow computation. After grid generation, the minimum geometric distances to each surface element are calculated and stored for each FP before the flow computation. The value d_{IP} is user-defined. We call the above boundary condition LO, an abbreviation of "low-order." LO is a first-order method.

In the LO method, IP must be in the fluid cells. To set d_{IP} longer than the diagonal line of a wall cell, the following condition must be satisfied:

$$d_{IP} \geq \sqrt{n_{DIM}} \Delta x, \quad (17)$$

where n_{DIM} denotes the dimension of the simulation. In this study, when LO is used as a boundary condition, d_{IP} is set to $3\Delta x$, as in [30].

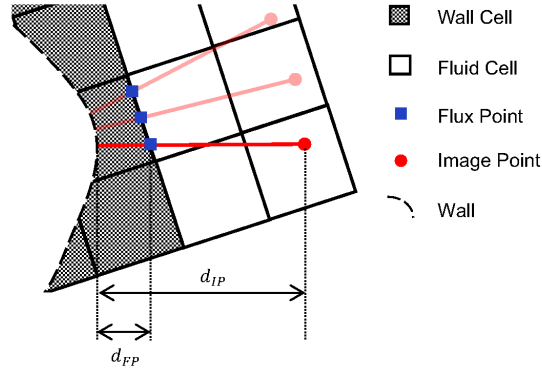
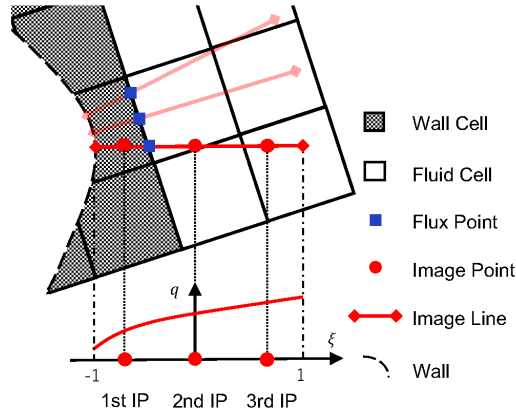


Fig. 6: Schematic of the boundary condition with a single IP.

Fig. 7: Schematic of the boundary condition which interpolates all the variables ($k = 3$).

2.3.2. High-order methods

Zero-order or first-order interpolation of physical quantities for a wall surface, such as a method that uses only one IP for each FP in Section 2.3.1 is known to be inaccurate at the near-wall region [20]. Therefore, we propose methods that use multiple IPs to calculate the physical quantities at the FPs with high-order accuracy using Eqs. (15). We use Eqs. (15), because the wall curvature appears in only one equation instead of two in Eqs. (14).

First, we describe the method for interpolating the four physical quantities from Eqs. (15). Figure 7 shows the schematic of the wall boundary condition at $k = 3$. A red line (Image-Line, IL) perpendicular to the wall is drawn across the FP (blue square), with the endpoint of IL on the wall as $\xi = -1$ and on the fluid side as $\xi = 1$. IPs are placed on each IL at Gauss point locations. Gauss points are used to generate the ξ -coordinates of IPs on ILs the same as to the ξ -coordinates of SPs in the cells, thereby simplifying the code implementation. The restrictions on IP placement

are described below. Only IPs on the IL through one of the FPs are shown, but in practice, IPs are placed on interfaces between the fluid and wall cells for all FPs. After placing the ILs and IPs, we construct the $(k - 1)$ th-order polynomial distributions of the physical quantities on ILs using the wall condition expressed by Eqs. (15) and the information of the IPs. Next, we interpolate the physical quantities at the FPs from these polynomials and compute the numerical fluxes at FPs using SLAU.

First, the physical quantity at the second to k th IP from the wall is obtained from the polynomial distribution in the fluid cell where each IP exists, using the Eq. (12). Then, following Eq. (3), a polynomial expansion for the Eqs. (15) is performed at the point on the wall ($\xi = -1$) as

$$\phi_1(-1)u_{n,1} + \sum_{i=2}^k \phi_i(-1)u_{n,i} = 0, \quad (18a)$$

$$\phi'_1(-1)H_1 + \sum_{i=2}^k \phi'_i(-1)H_i = 0, \quad (18b)$$

$$\phi'_1(-1)S_1 + \sum_{i=2}^k \phi'_i(-1)S_i = 0, \quad (18c)$$

$$\frac{2}{d_{IL}} \left\{ \phi'_1(-1)u_{t,1} + \sum_{i=2}^k \phi'_i(-1)u_{t,i} \right\} = \frac{\phi_1(-1)u_{t,1} + \sum_{i=2}^k \phi_i(-1)u_{t,i}}{R}, \quad (18d)$$

where the subscript number of each physical quantity denotes the IP index from the wall, and d_{IL} denotes the length of the IL. Solving Eq. (18) for the first IP from the wall, we obtain the following equations:

$$u_{n,1} = -\frac{\sum_{i=2}^k \phi_i(-1)u_{n,i}}{\phi_1(-1)}, \quad (19a)$$

$$H_1 = -\frac{\sum_{i=2}^k \phi'_i(-1)H_i}{\phi'_1(-1)}, \quad (19b)$$

$$S_1 = -\frac{\sum_{i=2}^k \phi'_i(-1)S_i}{\phi'_1(-1)}, \quad (19c)$$

$$u_{t,1} = -\left(\frac{1}{R} \sum_{i=2}^k u_{t,i} \phi_i(-1) - \frac{2}{d_{IL}} \sum_{i=2}^k u_{t,i} \phi'_i(-1) \right) \left/ \left(\frac{1}{R} \phi_1(-1) - \frac{2}{d_{IL}} \phi'_1(-1) \right) \right. \quad (19d)$$

As a result, the physical quantities for all IPs on the IL are obtained, and polynomial distributions on the IL can be constructed. We obtain the u_n, H, S, u_t at the FP from the polynomial distributions and transform them into primitive variables. Finally, we compute the numerical fluxes at the FP using SLAU. We call the above boundary condition HO-i/c. "HO" denotes "high-order," "i" denotes interpolating u_t , and "c" represents "curvature."

The wall boundary condition that approximates the radius of curvature in Eq. (19d) to infinity is called HO-i/f. Here, "f" denotes "flat," implying that this method approximates the wall as flat. In this case, the tangential component of the velocity at the first IP from the wall, $u_{t,1}$, is given by

$$u_{t,1} = -\frac{\sum_{i=2}^k \phi'_i(-1)u_{t,i}}{\phi'_1(-1)}. \quad (20)$$

Next, we will examine the high-order method that does not employ wall curvature. The wall curvature must be calculated for HO-i/c, as described previously. However, it can be challenging to compute the curvature of complex

geometries robustly and accurately. Therefore, we extrapolate u_t from the fluid cell without using Eq. (15d) which includes the wall curvature. (15d).

Figure 8 shows the schematic of the wall boundary conditions at $k = 3$. We place Image Points (Image Points A, IP_A) at the same position as HO-i/c and HO-i/f. Next, we place other Image Points (Image Points B, IP_B) on the coordinates where $\eta = -1$ is placed at the FP position and $\eta = 1$ at the same position as $\xi = 1$. This method uses IP_A to interpolate u_n, H, S in the same way as HO-i/c and HO-i/f, and IP_B to extrapolate u_t . Here, u_t are extrapolated to FP using only fluid cell information as follows:

$$u_{t,FP} = \sum_{i=1}^k \phi_i(-1)u_{t,i}^B. \quad (21)$$

The superscript B on the physical quantity indicates a physical quantity of IP_B . The above method is called HO-e, where "e" means extrapolating u_t .

For HO-i/c and HO-i/f, the second IP from the wall, and for HO-e, the second IP_A from the wall must be in the fluid cells, so the blue line in Figure 9 must be longer than a diagonal line of the wall cell. The following condition must be satisfied:

$$d_{IL} \frac{1 + \xi_2}{2} \geq \sqrt{n_{DIM}} \Delta x, \quad (22)$$

where ξ_2 denotes the ξ -coordinate of the second IP or IP_A from the wall on the IL. This condition determines the lower limit of d_{IL} . As obtained from Eq. (22), the lower limits are $1.79\Delta x$, $2.83\Delta x$, and $4.29\Delta x$ for $k = 2, 3$, and 4 , for 2-D simulations, and $2.20\Delta x$, $3.46\Delta x$, and $5.25\Delta x$ for 3-D simulations, respectively. On the other hand, the length of the d_{IL} should not be unnecessarily long. Thus, when using HO-i/c, HO-i/f, and HO-e, we let d_{IL} be $3\Delta x$, $3\Delta x$, and $5\Delta x$ for $k = 2, 3$, and 4 , for 2-D simulations, and $3\Delta x$, $5\Delta x$, and $7\Delta x$ for 3-D simulations, respectively.

When the IP or IL ends in a cell containing a portion of another body, we do not use immersed boundary method. To avoid a numerical problem, the wall boundary is considered as a step-wise face, and the values at FPs on the interfaces between fluid and wall cells are determined using the fluid cells including the FPs.

2.3.3. Extension to 3-D Simulations

The wall boundary conditions described above (LO, HO-i/c, HO-i/f, and HO-e) are extended to 3-D simulations. For LO, the wall boundary conditions are imposed using the same operations as for 2-D simulations. For HO, interpolation using the Eqs. (15) is performed in the plane formed by the unit vector s representing the tangential direction to the wall and the unit vector n representing the perpendicular direction to the wall. The vector s is determined using the velocity component tangential to the wall u_s as:

$$s = \frac{u_s}{\|u_s\|}, u_s = u_2 - (u_2 \cdot n)n, \quad (23)$$

where u_2 is the velocity vector at second IP or IP_A from the wall.

2.4. Time integration

The explicit time integration method is the four-stage, fourth-order Runge-Kutta method, while the implicit time integration method is MFGS [26]. In the conventional finite volume method, the Jacobi matrix on the left-hand side

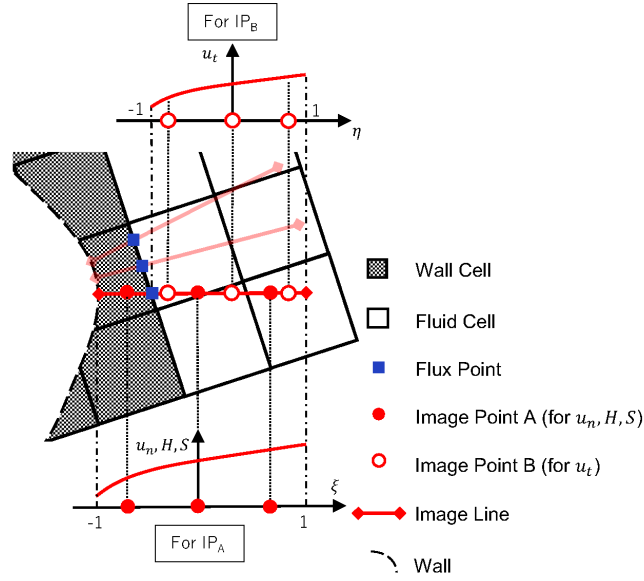


Fig. 8: Schematic of the boundary condition which extrapolates the velocity component tangential to the wall u_t ($k = 3$).

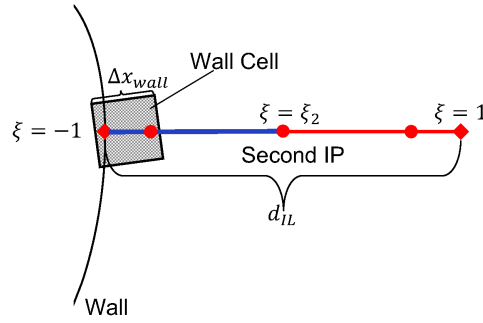


Fig. 9: Schematic of the condition of d_{IL} . The blue line should be longer than the diagonal line of the wall cell.

of the implicit method is sparse. Due to internal degrees of freedom, the FR method yields a dense matrix, which is computationally expensive [31]. Therefore, defect correction on sub-cells (DECS) [31] is used to reduce the computational cost. We consider small cells (sub-cells) that include each SP in a cell and discretize the sub-cells using first-order FVM.

3. Numerical tests

3.1. 2-D inviscid vortex propagation

We analyze an inviscid vortex advection problem [32] to validate the accuracy of the inviscid flow simulations without the wall. The following are the initial values of the physical quantities:

$$\rho = 1, \quad (24a)$$

$$u = M_\infty \left\{ 1 - \kappa y \exp\left(-\frac{r^2}{2}\right) \right\}, \quad (24b)$$

$$v = M_\infty \kappa x \exp\left(-\frac{r^2}{2}\right), \quad (24c)$$

$$P = \frac{1}{\gamma} - \frac{1}{2} \kappa^2 M_\infty^2 \rho \exp(-r^2). \quad (24d)$$

The above physical quantities are dimensionless, and $r = \sqrt{x^2 + y^2}$, $\kappa = 0.02$, M_∞ is the freestream Mach number, set to 0.1. Because this is an inviscid simulation, the ideal solution is one in which a uniform flow advects the vortex represented by the initial value. The direction of the advection is along the x -axis. The computational domain is a square with $-6 \leq x \leq 18$ and $-12 \leq y \leq 12$. Periodic boundary conditions are imposed on the boundary in the x -direction and Riemann boundary conditions in the y -direction. Uniform grids with minimum grid widths Δx_{min} of 0.375, 0.1875, and 0.09375 are employed. The analysis is performed with k , where the number of SPs are 2, 3, and 4. For the time integration, we use a four-stage fourth-order Runge-Kutta method with a sufficiently small time step such that further reduction of the time step does not affect the results. The L_∞ error in the y -directional velocity when the vortex is at $x = 8$ is compared with the theoretical solution.

Figure 10 shows the results of the L_∞ error in y -directional velocity. The horizontal axis is the square root of the degrees of freedom, calculated as

$$NDoF = NGrid \times k^2, \quad (25)$$

where $NGrid$ denotes the total number of grids. Compared to the dashed lines in the figure that show the second, third, and fourth-order accuracy slopes, the results of all simulations show slopes corresponding to the order. The constructed solver achieves k th-order accuracy for the simulation of inviscid flows without the wall.

3.2. 2-D inviscid flow through a channel with a bump

We evaluate the model accuracy with a smooth bump flow through a channel. The freestream has a Mach number of 0.5. The computational domain is set from -1.5 to 1.5 in the x -direction, the upper boundary in the y -direction is $y = 0.8$, and the lower boundary is the bump expressed as follows:

$$y = 0.0625 \exp(-25x^2). \quad (26)$$

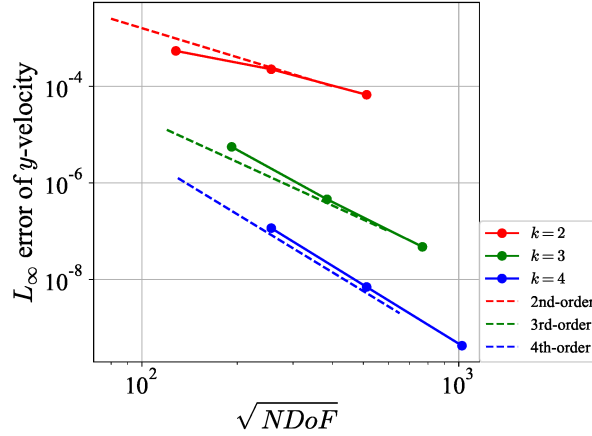


Fig. 10: L_∞ error of y -velocity in inviscid vortex propagation.

At the $x = -1.5$ inflow boundary, the total temperature T_0 and total pressure P_0 are defined to satisfy the following conditions:

$$\frac{T_0}{T} = \left(1 + \frac{\gamma - 1}{2} M^2\right), \quad (27a)$$

$$\frac{P_0}{P} = \left(\frac{T_0}{T}\right)^{\frac{\gamma}{\gamma-1}}. \quad (27b)$$

The outflow boundary at $x=1.5$ is subjected to a constant static pressure condition. The upper and lower boundaries are slip walls, and the four boundary conditions are evaluated: LO, HO-i/c, HO-i/f, and HO-e. The radius of the wall curvature used in HO-i/c is obtained analytically from the Eq. (26). Uniform grids are utilized as computational grids. The minimum grid widths Δx_{min} are 0.0250, 0.0168, 0.0100, and 0.00625. Figure 11 shows an example of the computational grids. We conduct steady flow simulation and use MFGS with local time stepping for time integration. We verify that the residuals are converged in all cases. As a measure of accuracy, we use the following L_2 error of entropy proposed by Wang et al. [33]:

$$Error_{L_2(\Omega)} = \sqrt{\frac{\int_{\Omega} \left\{ \frac{P}{P_{\infty}} \left(\frac{\rho}{\rho_{\infty}} \right)^{\gamma} - 1 \right\}^2 dV}{V}}, \quad (28)$$

where ρ_{∞} and P_{∞} denote the density and pressure of the uniform flow, V denotes the volume, and Ω denotes the entire computational domain.

Figure 12 demonstrates the L_2 error of entropy. The horizontal axis is the square root of the degrees of freedom expressed as Eq. (25). Compared to the dashed lines in the figure, which show the second, third, and fourth-order accuracy slopes, the HO-i/c and HO-e results show slopes that correspond to the order. HO-i/c expresses Eqs. (15) accurately by interpolating the four physical quantities with high-order polynomials while considering the wall curvature. HO-e achieves the same order accuracy as the FR method without curvature by extrapolating u_t using only information from fluid cells within the domain. On the other hand, the LO and HO-i/f methods produce lower-order results than the FR method. This is because LO interpolates physical quantities with first-order accuracy, whereas

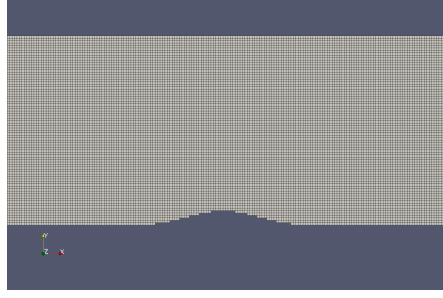


Fig. 11: Computational grid for inviscid flow through a channel with a bump ($\Delta x_{min} = 0.0100$).

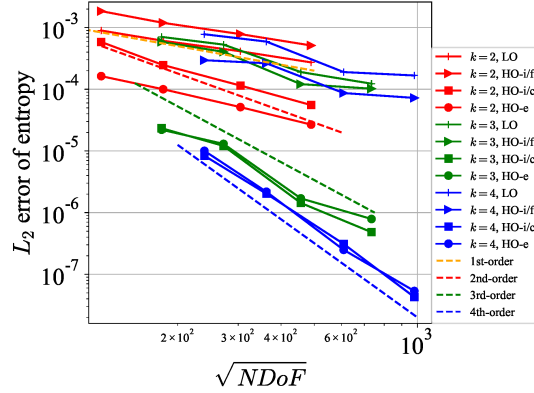


Fig. 12: L_2 error of entropy in 2-D inviscid flow through a channel with a bump.

HO-i/f approximates the wall curvature to infinity, although the degree of the polynomial on IL is $k - 1$. In terms of error values, HO-i/c and HO-e are one to three orders of magnitude less than LO and HO-i/f. Figure 13 shows the stream line and Mach number distribution at $k = 3, \Delta x_{min} = 0.0100$, and Figure 15 shows the pressure coefficient distribution at $k = 3, \Delta x_{min} = 0.0100$. For each boundary condition, the distributions are nearly identical. We compare the distribution of physical quantities on the surface to see how each boundary condition differs. Figure 15 shows the difference in surface pressure coefficient upstream and downstream of the apex of the bump in the same case. To compute the pressure coefficient on the surface, wall normal lines are set through the gravity center of each surface element, and the pressure coefficient are extrapolated to the element with the same polynomial order as the simulation. The vertical axis ΔC_p is defined as follows:

$$\Delta C_p(|x|) = C_p(-|x|) - C_p(|x|), \quad (29)$$

where $|x|$ denotes the absolute value of the x -coordinate with the apex of the bump as the origin. Therefore, the ΔC_p is zero if the flow is symmetric in reference to the apex. The ΔC_p of HO-i/c and HO-e are smaller than LO and HO-i/f, and the flows in HO-i/c and HO-e show more line-symmetrical patterns about the y -axis through the apex of the bump. Furthermore, Table 1 shows the drag coefficients. To compute the drag coefficients, we integrate numerical fluxes on each FP on the interfaces between fluid and wall cells [10]. Drag coefficients in inviscid flow simulations should ideally be zero. From Table 1, the drag coefficients of the HO-i/c and HO-e are closer to zero than the other

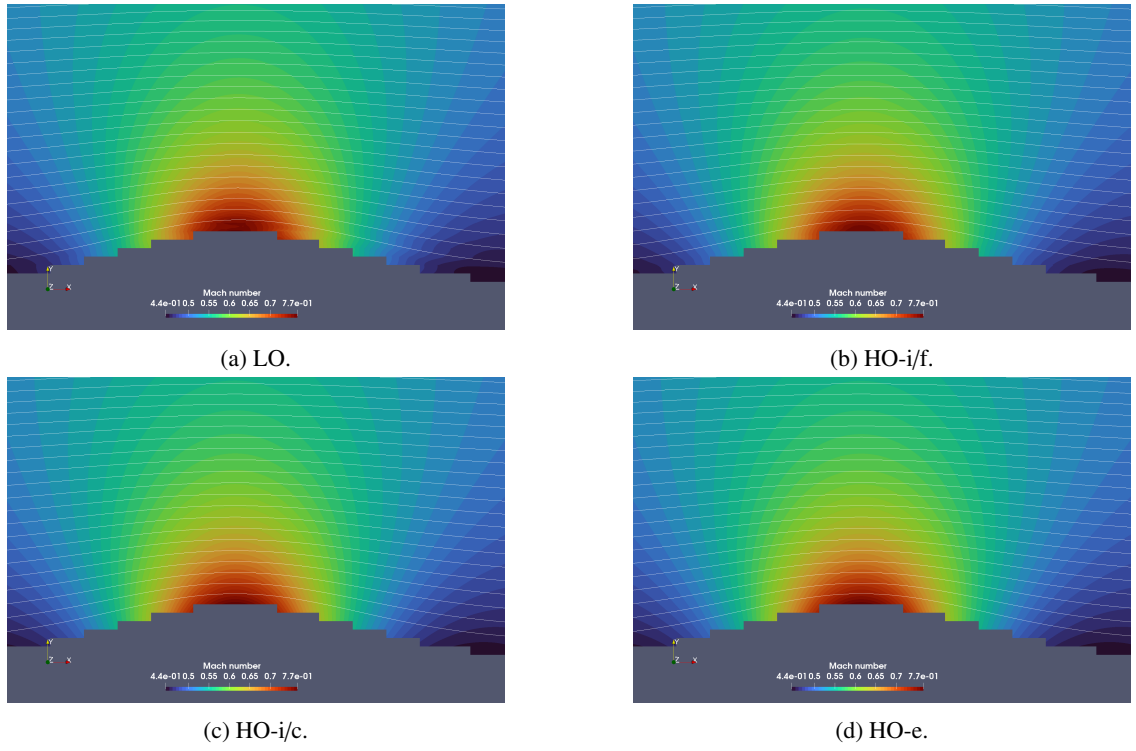


Fig. 13: The stream line and Mach number distribution in 2-D inviscid flow through a channel with a bump ($k = 3, \Delta x_{min} = 0.0100$).

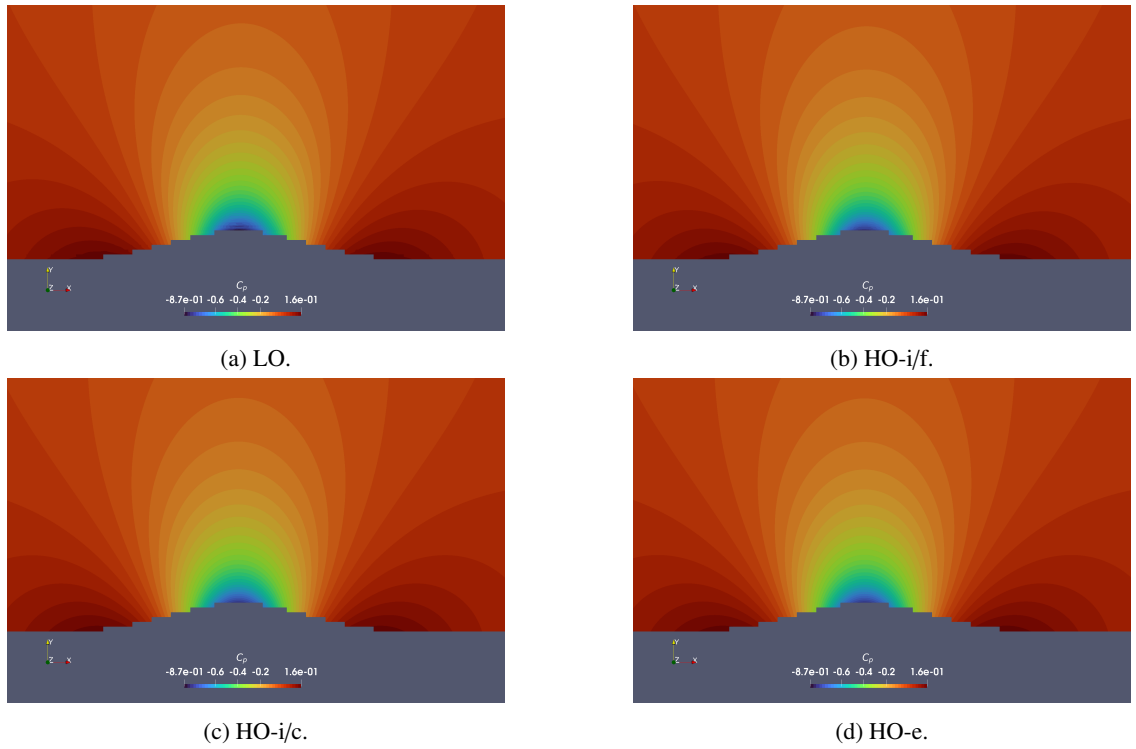


Fig. 14: Pressure coefficient distribution in 2-D inviscid flow through a channel with a bump ($k = 3, \Delta x_{min} = 0.0100$).

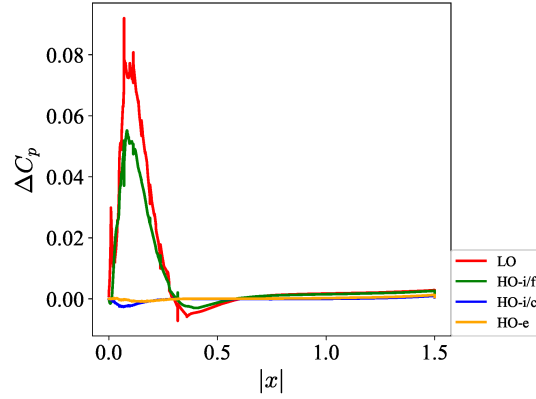


Fig. 15: The difference in surface pressure coefficient upstream and downstream of the apex of the bump ($k = 3$, $\Delta x_{min} = 0.0100$).

two methods.

3.3. 2-D inviscid flow past a cylinder

We evaluate the model accuracy with an inviscid flow around a cylinder as a more general geometry analysis. The Mach number is equal to 0.3. The computational domain is a 200×200 square area, with a diameter of one. We employ non-uniform grids and specify minimum grid widths Δx_{min} to 0.0977, 0.0488, and 0.0244. Figure 16 shows the computational grid for $\Delta x_{min} = 0.0244$. The Riemann boundary condition is used as the far-field boundary condition, and four types of wall boundary conditions are tested: LO, HO-i/c, HO-i/f, and HO-e. The radius of the cylinder is the radius of the wall curvature used in HO-i/c. We conduct steady flow simulation and use MFGS with local time stepping for time integration. We verify that the residuals are converged in all cases. Because entropy is ideally conserved, the accuracy verification indicator is the entropy L_2 error expressed as Eq. (28).

Figure 17 shows a graph with the horizontal axis representing the square root of the degrees of freedom and the vertical axis representing the L_2 error in entropy. The results of all simulations show a much steeper slope compared to the dashed lines in the figure. This is due to the inappropriate selection of the horizontal axis. Figure 18 shows the distribution of L_2 entropy error calculated in each cell at $k = 3$, $\Delta x_{min} = 0.0244$, with the boundary condition being HO-i/c. The regions with significant L_2 entropy errors are concentrated near the wall, and the grid's resolution near the wall significantly affects the error magnitude. Thus, in this case, we use the inverse of the minimum grid width as the horizontal axis.

Figure 19 demonstrates a graph where the horizontal axis is the inverse of the minimum grid width, and the vertical axis is the L_2 error of entropy. LO and HO-i/f are incapable of achieving the k th-order accuracy. However, HO-i/c, which considers the wall curvature, and HO-e, which extrapolates u_t from the fluid cell, achieve k th-order accuracy. This result is consistent with the bump flow results in Section 3.2. Figure 20 shows the streamline and the spatial distribution of the velocity component in the x -direction at $k = 3$, $\Delta x_{min} = 0.0244$. For LO and HO-i/f, the velocity distribution is asymmetrical to a vertical line passing through the center of the cylinder. On the other hand, the velocity distribution is close to line-symmetric. In inviscid flow simulations, drag coefficients are ideally zero.

Table 1: Drag coefficients in 2-D inviscid flow through a channel with a bump ($k = 3$, $\Delta x_{min} = 0.0100$).

Boundary condition	C_d
LO	2.7×10^{-5}
HO-i/f	3.6×10^{-5}
HO-i/c	7.8×10^{-7}
HO-e	1.0×10^{-7}

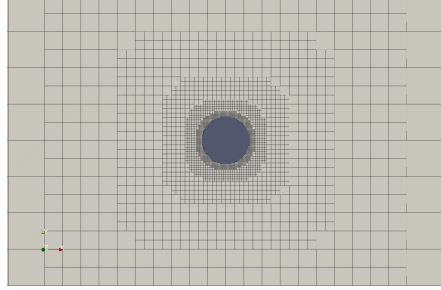
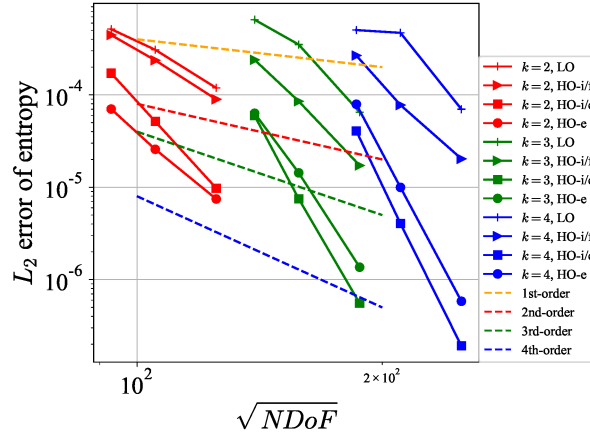
Fig. 16: Computational grid for the flow around a cylinder (near the wall, $\Delta x_{min} = 0.0244$).Fig. 17: L_2 error of entropy in the flow around a cylinder (the horizontal axis is the square root of the degrees of freedom).

Table 2 shows that the drag coefficients of HO-i/c and HO-e are closer to zero than the other two. Figure 21 shows the distribution of pressure coefficients on the wall surface. Here, the theoretical value of the pressure coefficient at the stagnation point is 1.023. HO-i/c and HO-e have pressure coefficients at the stagnation point ($\theta = \pi$) and the rear stagnation point ($\theta = 0$), closer to the theoretical value than the other two methods.

3.4. 3-D inviscid flow past a sphere

For the 3-D case, we simulate inviscid flow around a sphere, with a Mach number of 0.2. The computational domain is -100 to 100 in the x -direction and -50 to 50 in the y and z directions. The flow direction is in the x -direction. The minimum grid width Δx_{min} is set to 0.0977, 0.0488, 0.0244, and 0.0122. Figure 22 demonstrates the computational grid for $\Delta x_{min} = 0.0122$. The far-field boundary condition is the Riemann boundary condition, and

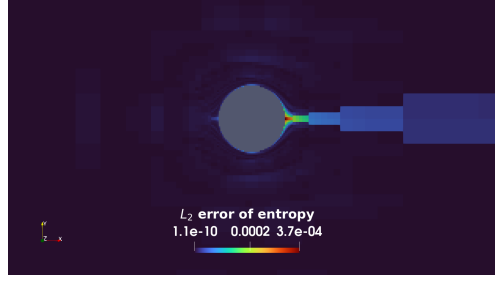


Fig. 18: The distribution of L_2 entropy error calculated in each cell ($\Delta x_{min} = 0.0244$, $k = 3$, HO-i/c).

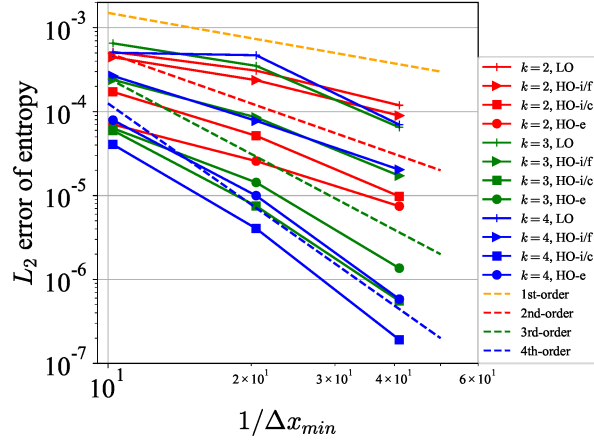
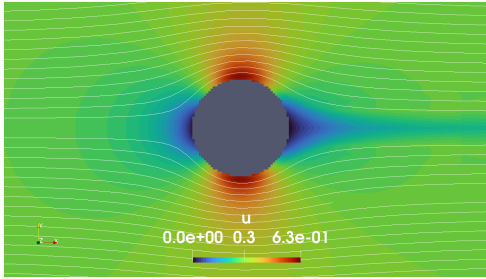
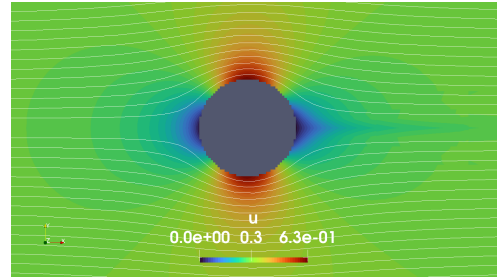


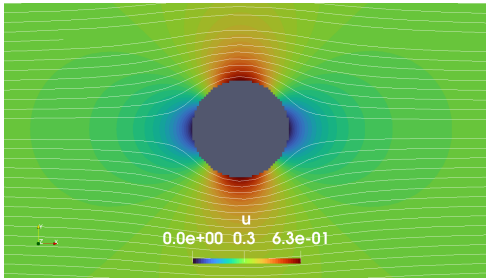
Fig. 19: L_2 error of entropy in the flow around a cylinder (the horizontal axis is the inverse of the minimum grid width).



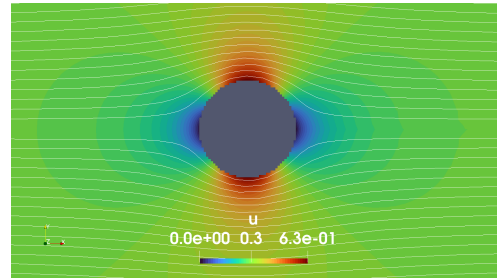
(a) LO.



(b) HO-i/f.



(c) HO-i/c.

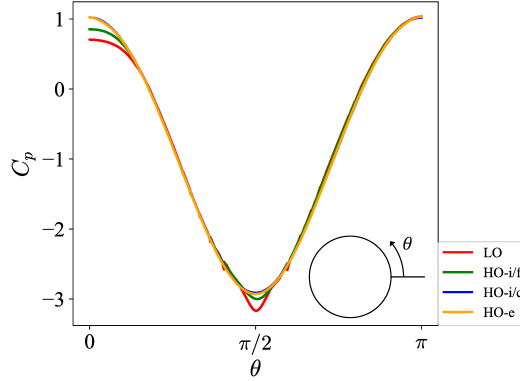
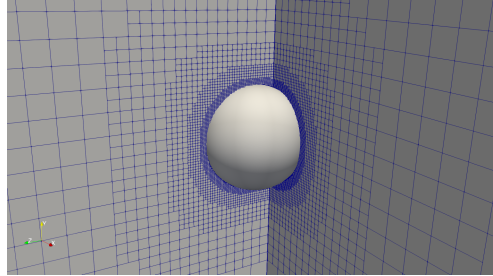


(d) HO-e.

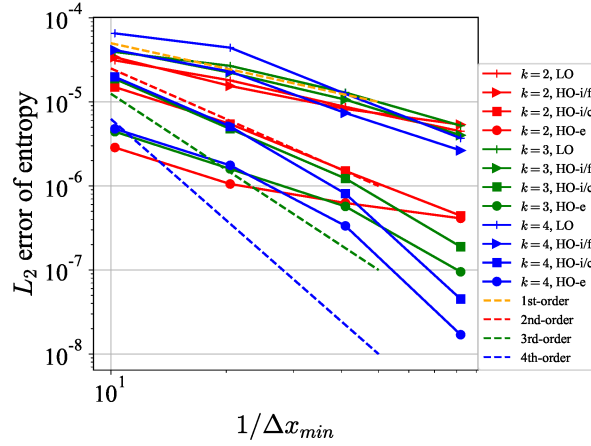
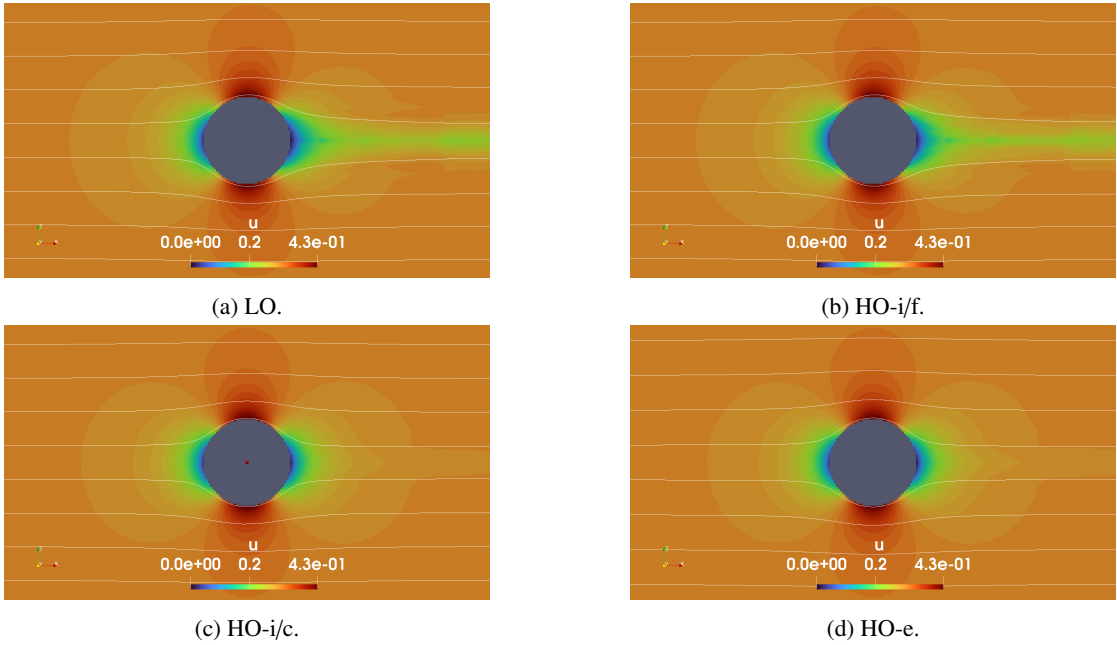
Fig. 20: The stream line and the distribution of the velocity component in x direction in 2-D inviscid flow around a cylinder ($k = 3$, $\Delta x_{min} = 0.0244$).

Table 2: Drag coefficients in 2-D inviscid flow past a cylinder ($k = 3, \Delta x_{min} = 0.0244$).

Boundary condition	C_d
LO	5.0×10^{-2}
HO-i/f	1.2×10^{-2}
HO-i/c	9.1×10^{-5}
HO-e	4.5×10^{-4}

Fig. 21: Surface pressure coefficients in 2-D inviscid flow past a cylinder ($k = 3, \Delta x_{min} = 0.0244$).Fig. 22: Computational grid for the flow around a sphere (near the wall, $\Delta x_{min} = 0.0122$).

four types of wall boundary conditions are used: LO, HO-i/c, HO-i/f, and HO-e. The wall curvature used in HO-i/c is the sphere's radius. The flow simulation is conducted under steady flow conditions using MFGS with local time stepping for time integration. We confirm the convergence of the residuals for all cases. Since entropy is conserved, the error verification indicator is the L_2 error of entropy in Eq. (28). Figure 23 shows a graph wherein the horizontal axis is the inverse of the minimum grid width and the vertical axis is the L_2 error of entropy. Here, LO and HO-i/f were not k th-order accurate. On the other hand, HO-i/c, which takes the wall curvature into account, and HO-e, which extrapolates u_t from the fluid cell, are k th-order accurate. This result is consistent with the bump flow results in Section 3.2 and the flow around a cylinder in Section 3.3. Figure 24 shows the streamline and the spatial distribution of the velocity component in the x -direction on the $y = 0$ plane at $k = 3, \Delta x_{min} = 0.0244$. As with the cylinder case, the velocity distribution is asymmetrical to the $x = 0$ plane for LO, HO-i/f, and close to symmetric for HO-i/c and HO-e. Table 3 shows that the drag coefficients of HO-i/c and HO-e are closer to zero than the other two methods. In

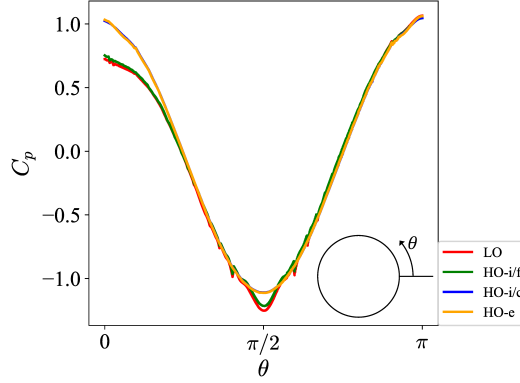
Fig. 23: L_2 error of entropy in the flow around a sphere.Fig. 24: The stream line and the distribution of the velocity component in x direction on $y = 0$ plane in 3-D inviscid flow past a sphere ($k = 3$, $\Delta x_{min} = 0.0244$).

addition, Figure 25 shows the distributions of the surface pressure coefficients on the $y = 0$ plane; HO-i/c and HO-e have pressure coefficients at the stagnation point ($\theta = \pi$) and the rear stagnation point ($\theta = 0$), closer to the theoretical value (1.010) than the other two methods.

As shown above, high-order accuracy can be achieved using the boundary conditions HO-i/c and HO-e. Of these two methods, HO-e is more versatile in terms of a robust and accurate analysis of objects with complex geometries since it does not require wall curvature.

Table 3: Drag coefficients in 3-D inviscid flow past a sphere ($k = 3, \Delta x_{min} = 0.0244$).

Boundary condition	C_d
LO	1.5×10^{-2}
HO-i/f	1.4×10^{-2}
HO-i/c	7.0×10^{-4}
HO-e	9.0×10^{-4}

Fig. 25: Surface pressure coefficients on $y = 0$ plane in 3-D inviscid flow past a sphere ($k = 3, \Delta x_{min} = 0.0244$).

4. Conclusion

In this paper, we proposed novel immersed boundary methods for inviscid flow simulations based on the discrete forcing approach to developing a hierarchical Cartesian grid flow solver based on the FR method, affording high-order accuracy analysis even in the near-wall region. Four methods are proposed: LO, which interpolates physical quantities using a single IP in a fluid cell; HO-i/c, HO-i/f, and HO-e, which use multiple IPs in fluid cells. Herein, LO interpolates physical quantities with first-order. HO-i/c, HO-i/f, and HO-e interpolate physical quantities by constructing higher-order polynomials on ILs with multiple IPs on the ILs. These methods leverage the FR method's ability to obtain physical quantities at arbitrary points from the polynomial distribution in a fluid cell. These methods apply to both 2-D and 3-D simulations.

We validated the solver's accuracy using these four boundary conditions and a fundamental inviscid flow. For the 2-D vortex advection problem (flow without wall), we demonstrated that the solver's accuracy is adequate for inviscid flow. We then validated the solver for flow fields with walls, including flow through a channel with a bump, flow around a cylinder, and flow around a sphere. Among the methods for polynomial interpolation of physical quantities on IL, the results demonstrated that HO-i/c and HO-e achieve accuracy comparable to the order of the FR method. HO-i/c employs the wall curvature and imposes the wall boundary conditions explicitly. HO-e extrapolates the velocity component tangential to the wall u_t inside the domain from the fluid cell to avoid using wall curvature. Of these two methods, HO-e is more versatile in terms of a robust and accurate analysis of objects with complex geometries since it does not require wall curvature.

References

- [1] J. Slotnick, A. Khodadoust, J. Alonso, D. Darmofal, W. Gropp, E. Lurie, D. Mavriplis, CFD Vision 2030 Study: A Path to Revolutionary Computational Aerosciences (2013) 58.
- [2] J. P. Slotnick, D. Mavriplis, A Grand Challenge for the Advancement of Numerical Prediction of High Lift Aerodynamics, in: AIAA Scitech 2021 Forum, American Institute of Aeronautics and Astronautics, VIRTUAL EVENT, 2021. doi:[10.2514/6.2021-0955](https://doi.org/10.2514/6.2021-0955).
- [3] B. Cockburn, G. E. Karniadakis, C.-W. Shu, The Development of Discontinuous Galerkin Methods, in: B. Cockburn, G. E. Karniadakis, C.-W. Shu (Eds.), Discontinuous Galerkin Methods, Lecture Notes in Computational Science and Engineering, Springer, Berlin, Heidelberg, 2000, pp. 3–50. doi:[10.1007/978-3-642-59721-3_1](https://doi.org/10.1007/978-3-642-59721-3_1).
- [4] Y. Liu, M. Vinokur, Z. Wang, Discontinuous Spectral Difference Method for Conservation Laws on Unstructured Grids, Journal of Computational Physics 216 (2004).
- [5] H. T. Huynh, A Flux Reconstruction Approach to High-Order Schemes Including Discontinuous Galerkin Methods, in: 18th AIAA Computational Fluid Dynamics Conference, American Institute of Aeronautics and Astronautics, Miami, Florida, 2007. doi:[10.2514/6.2007-4079](https://doi.org/10.2514/6.2007-4079).
- [6] M. J. Berger, M. J. Aftosmis, An ODE-Based Wall Model for Turbulent Flow Simulations, AIAA Journal 56 (2018) 700–714.
- [7] F. Capizzano, Turbulent Wall Model for Immersed Boundary Methods, AIAA Journal 49 (2011) 2367–2381.
- [8] K. Sugaya, T. Imamura, Turbulent flow simulations of the common research model on Cartesian grids using recursive fitting approach, Journal of Computational Physics 467 (2022) 111460.
- [9] T. Imamura, Y. Tamaki, Unsteady Flow Simulation around Two-Wheel Main Landing Gear based on Compressible Navier-Stokes Solver with Immersed Boundary Method, in: AIAA AVIATION 2020 FORUM, American Institute of Aeronautics and Astronautics, VIRTUAL EVENT, 2020. doi:[10.2514/6.2020-2579](https://doi.org/10.2514/6.2020-2579).
- [10] Y. Tamaki, T. Imamura, Turbulent Flow Simulations of the Common Research Model Using Immersed Boundary Method, AIAA Journal 56 (2018) 2271–2282.
- [11] C. S. Peskin, Numerical analysis of blood flow in the heart, Journal of Computational Physics 25 (1977) 220–252.
- [12] R. M. G. Iaccarino, IMMERSED BOUNDARY METHODS, Annu. Rev. Fluid Mech (2004) 24.
- [13] C. S. Peskin, Flow patterns around heart valves: A numerical method, Journal of Computational Physics 10 (1972) 252–271.
- [14] R. P. Beyer, A computational model of the cochlea using the immersed boundary method, Journal of Computational Physics 98 (1992) 145–162.
- [15] J. Kou, S. Joshi, A. Hurtado-de-Mendoza, K. Puri, C. Hirsch, E. Ferrer, Immersed boundary method for high-order flux reconstruction based on volume penalization, Journal of Computational Physics (2021) 110721.
- [16] P. De Palma, M. de Tullio, G. Pascasio, M. Napolitano, An immersed-boundary method for compressible viscous flows, Computers & Fluids 35 (2006) 693–702.
- [17] M. de Tullio, P. De Palma, G. Iaccarino, G. Pascasio, M. Napolitano, An immersed boundary method for compressible flows using local grid refinement, Journal of Computational Physics 225 (2007) 2098–2117.
- [18] D. Appellö, N. A. Petersson, A Fourth-Order Accurate Embedded Boundary Method for the Wave Equation, SIAM Journal on Scientific Computing 34 (2012) A2982–A3008.
- [19] A. Baeza, P. Mulet, D. Zorío, High Order Boundary Extrapolation Technique for Finite Difference Methods on Complex Domains with Cartesian Meshes, Journal of Scientific Computing 66 (2016) 761–791.
- [20] J. H. Seo, R. Mittal, A high-order immersed boundary method for acoustic wave scattering and low-Mach number flow-induced sound in complex geometries, Journal of Computational Physics 230 (2011) 1000–1019.
- [21] S. Clain, D. Lopes, R. Pereira, Very high-order Cartesian-grid finite difference method on arbitrary geometries, Journal of Computational Physics 434 (2021) 110217.
- [22] M. E. Khalili, M. Larsson, B. Müller, High-order ghost-point immersed boundary method for viscous compressible flows based on summation-by-parts operators: High-order immersed boundary method for viscous compressible flows, International Journal for Numerical Methods in Fluids 89 (2019) 256–282.
- [23] A. Dadone, B. Grossman, Surface boundary conditions for the numerical solution of the Euler equations, AIAA Journal 32 (1994) 285–293.
- [24] J. Fernández-Fidalgo, S. Clain, L. Ramírez, I. Colominas, X. Nogueira, Very high-order method on immersed curved domains for finite difference schemes with regular Cartesian grids, Computer Methods in Applied Mechanics and Engineering 360 (2020) 112782.
- [25] Y. Zhang, C. H. Zhou, An immersed boundary method for simulation of inviscid compressible flows: AN IMMERSED BOUNDARY METHOD FOR INVISCID COMPRESSIBLE FLOWS, International Journal for Numerical Methods in Fluids 74 (2014) 775–793.
- [26] E. Shima, K. Kitamura, Parameter-Free Simple Low-Dissipation AUSM-Family Scheme for All Speeds, AIAA Journal 49 (2011) 1693–1709.
- [27] D. A. Kopriva, J. H. Kolas, A Conservative Staggered-Grid Chebyshev Multidomain Method for Compressible Flows, Journal of Computational Physics 125 (1996) 244–261.
- [28] N. Balakrishnan, G. Fernandez, Wall boundary conditions for inviscid compressible flows on unstructured meshes, International Journal for Numerical Methods in Fluids 28 (1998) 1481–1501.
- [29] A. Chertock, A. Coco, A. Kurganov, G. Russo, A Second-Order Finite-Difference Method for Compressible Fluids in Domains with Moving Boundaries, Communications in Computational Physics 23 (2018).
- [30] Y. Tamaki, M. Harada, T. Imamura, Near-Wall Modification of Spalart–Allmaras Turbulence Model for Immersed Boundary Method, AIAA Journal 55 (2017) 3027–3039.
- [31] K. Miyaji, H. Koyama, Efficient Implicit Method for a High-Order Flux Reconstruction Approach Using Modified Matrix-Free Defect Correction, Transactions of the Japan Society for Aeronautical and Space Sciences, Aerospace Technology Japan 20 (2022) 16–25.
- [32] S. Kawai, S. Lele, Localized artificial diffusivity scheme for discontinuity capturing on curvilinear meshes, Journal of Computational Physics 227 (2008) 9498–9526.
- [33] Z. Wang, K. Fidkowski, R. Abgrall, F. Bassi, D. Caraeni, A. Cary, H. Deconinck, R. Hartmann, K. Hillewaert, H. Huynh, N. Kroll, G. May, P.-O. Persson, B. van Leer, M. Visbal, High-order CFD methods: Current status and perspective: HIGH-ORDER CFD METHODS, International Journal for Numerical Methods in Fluids 72 (2013) 811–845.